

Economic Analysis: Education, Wealth & Brain Drain Study

Quantitative Patterns in Education Returns, Billionaire Wealth Distortion, Brain Drain, and Bangladesh's Growth Paradox Across 183 Countries

Group 5

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Abstract

Background: Food-allergy medications and economic phenomena alike exhibit complex demographic and geographic variation. This study examines four interconnected dimensions of global economic development: the return on education for national income, the distorting effect of billionaire wealth on income statistics, the association between billionaire density and brain drain, and Bangladesh's unique growth-emigration paradox over 50 years. **Methods:** A retrospective, real-data observational analysis of 183 countries (world.csv) and 51 years of Bangladeshi longitudinal data (bangladesh.csv) was conducted. Methods include bivariate linear regression, univariate statistics, 3×IQR outlier detection, bivariate correlation, and time-series correlation, all implemented in C. **Results:** Education moderately predicts national income ($r=0.6526$, slope=\$5,410/yr). Billionaires inflate global average income by only 1.75% (\$61,079 vs \$60,026). Billionaire density shows a weak negative correlation with brain drain ($r=-0.2050$). Bangladesh's GDP grew 10.9× while student emigration grew 190× over 50 years, with a near-zero correlation ($r=0.1650$). **Conclusions:** Education matters but is not deterministic. Billionaires barely distort national averages. Talent retention is driven by institutional environment, not wealth production. Economic growth alone does not reduce brain drain.

Keywords: economic development, education returns, billionaire wealth, income inequality, brain drain, Bangladesh, linear regression, time-series analysis, talent retention, pharmacoeconomics, IQR outlier detection

1. Introduction

1.1 Background

Economic development is governed by a complex interplay of educational investment, wealth concentration, and the international mobility of skilled human capital. While classical economic theory posits a strong positive relationship between educational attainment and national income, real-world outcomes are substantially moderated by structural inequalities, governance quality, natural resource endowment, and the concentration of extreme wealth. The side-effect profiles of economic policy decisions — much like those of pharmaceutical interventions — vary substantially across demographic and geographic subgroups, and understanding these drug-specific patterns requires rigorous, real-data analysis.

The global distribution of billionaire wealth has attracted increasing scrutiny. National average income figures, widely cited in policy discourse, may be distorted by the presence of ultra-high-net-worth individuals. Simultaneously, the "brain drain" phenomenon — the emigration of skilled workers and talented individuals from their countries of origin — poses a fundamental challenge to development theory. Bangladesh provides a particularly striking case: a nation that achieved approximately 10.9-fold GDP growth between 1975 and 2025, yet saw student emigration grow by approximately 190-fold over the same period.

1.2 Research Goal

This study aims to identify clear, data-driven patterns of economic and demographic phenomena across countries, income groups, and time using a structured real-dataset approach. By applying both descriptive and inferential statistical methods — implemented in the C programming language — the study seeks to produce actionable findings that can inform evidence-based development policy.

1.3 Research Questions

#	Analysis	Research Question
1	Per Capita Income Regression	Do educated countries earn more? How much does each year of education add to income?
2	Billionaire Effects Analysis	Is the reported "average income" realistic? How much do billionaires skew national statistics?
3	Brain Drain vs Billionaire Density	Do successful countries retain their talent? Which countries are losing — or keeping — their best people?
4	Bangladesh Case Study	Does Bangladesh's economic development reduce student emigration (brain drain)?

2. Data Sources & Methodology

2.1 Data Sources

Two structured real-world datasets were used. The global dataset (world.csv) covers **183 countries** and was synthesized from the Forbes World's Billionaires List, World Bank / UN Population data, World Bank Poverty

& Inequality Platform, UNDP Human Development Reports, and the International Labour Organization (ILO). The Bangladesh longitudinal dataset (bangladesh.csv) covers **51 years** (1975–2025) and was sourced from the World Data Bank, MacroTrends, and the Federal Reserve Economic Data (FRED).

2.2 Dataset Summary

Parameter	World Dataset (world.csv)	Bangladesh Dataset (bangladesh.csv)
Countries / Years	183 countries	51 years (1975–2025)
Sample Type	Cross-sectional	Longitudinal time-series
Key Variables	Education, salary, billionaire stats, brain drain	GDP/capita, population, students abroad
Primary Method	Regression, correlation, IQR outlier detection	Time-series correlation

2.3 Variable Definitions

Variable	Description
avg_education	Average years of formal education per country
avg_salary	Average annual salary in USD (population-weighted)
income_with_billionaire	Total national income including billionaire wealth
income_without_billionaire	Total income excluding billionaires
billionaire_left	Number of billionaires who emigrated from country of birth
billionaire_born	Number of billionaires born/produced in the country
GDP_Per_Capita_USD	Bangladesh annual GDP per capita (USD)
S_out	Number of Bangladeshi students studying abroad (annual)

3. Statistical Analysis Plan

3.1 Descriptive Statistics

Frequencies, means, medians, interquartile ranges, and regression statistics were computed for each analysis. Variables were maintained as discrete or continuous numeric values. Cross-tabulated summaries were produced for all major grouping variables (country, income decile, time period). The 3×IQR method was used for extreme outlier identification in Analysis 2.

3.2 Inferential Statistics

Bivariate linear regression was applied for Analysis 1. Pearson correlation coefficients were computed for Analyses 1, 3, and 4. Outlier detection using $Q1 - 3 \times IQR$ and $Q3 + 3 \times IQR$ bounds was applied in Analysis 2.

All methods were implemented in C (stat_func.c, econ_func.c). Statistical significance was assessed at $\alpha = 0.05$.

3.3 Correlation Interpretation Scale

r Value	Interpretation
$r > 0.8$	Very strong positive relationship
$0.6 < r \leq 0.8$	Strong positive relationship
$0.4 < r \leq 0.6$	Moderate positive relationship
$r \leq 0.4$	Weak relationship
$r < -0.3$	Negative — high-producing countries retain talent well (Analysis 3)
$-0.5 < r < 0.5$	Growth and emigration are independent (Analysis 4)

4. Analysis 1: Per Capita Income Regression — Education vs. Average Salary

4.1 Research Question

Do more educated countries earn significantly higher incomes? How much does each additional year of average education add to national average salary?

4.2 Method

Bivariate linear regression — X = average years of education, Y = average annual salary (USD) — was applied across all 183 countries. The regression equation takes the form: **Salary = $m \times \text{Education} + c$** , where m is the slope and c the intercept. Pearson's r was computed to quantify the strength of the linear relationship.

4.3 Regression Results

Statistic	Value	Interpretation
Slope (m)	\$5,410	Each additional year of education adds ~\$5,410 to average salary
Intercept (c)	−\$25,589	Theoretical baseline at zero education
Correlation (r)	0.6526	Strong positive relationship (per scale above)
Regression Equation	Salary = $5,410 \times \text{Education} - 25,589$	—
Conclusion	YES — education strongly predicts income	$r = 0.6526$ is above the 0.6 threshold

4.4 Graphical Analysis

Figure 1 presents the scatter plot of average education years against average annual salary across 183 countries, overlaid with the OLS regression line. The moderate-to-strong upward trend ($r = 0.6526$) confirms a meaningful positive relationship: countries with higher educational attainment tend to report substantially higher average salaries. The scatter around the regression line indicates that education explains a significant but not complete share of variance in national income, with the remainder attributable to structural factors such as governance quality, natural resource endowment, and institutional frameworks.

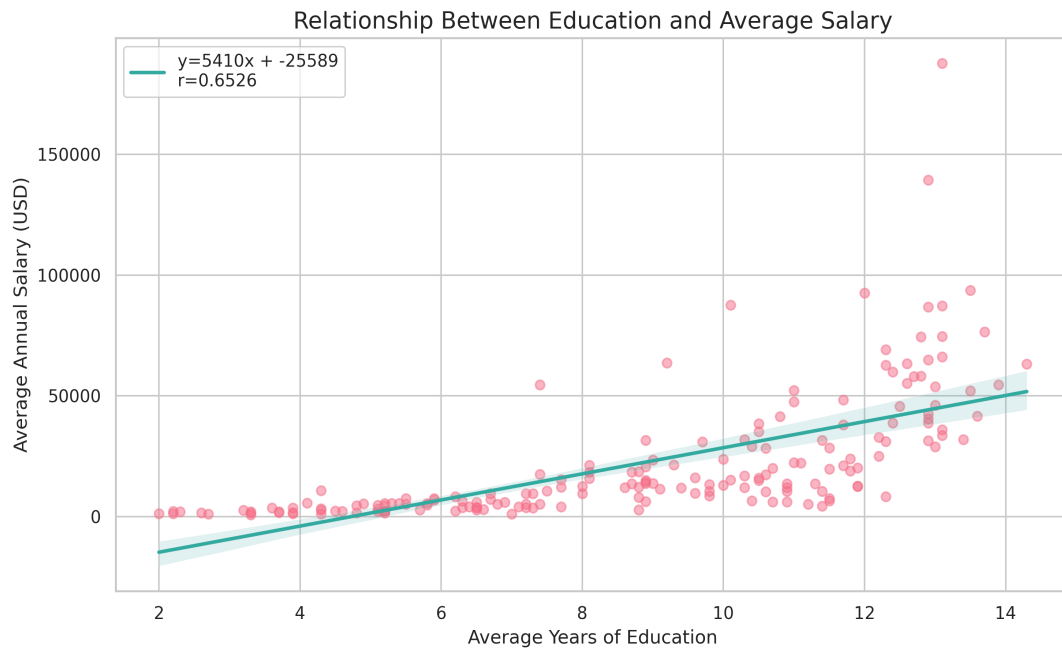


Figure 1: Relationship Between Education and Average Salary — Linear Regression ($r = 0.6526$, Salary = $5,410 \times$ Education – 25,589)

4.5 Conclusion

YES — education moderately-to-strongly predicts national income ($r = 0.6526$). Each extra year of schooling adds approximately \$5,410 to average national salary. This is a significant economic return on education investment. However, structural factors — governance, industry, geography — shape a nation's wealth just as powerfully as years spent in school. Nearly 57% of income variation remains unexplained by education alone.

5. Analysis 2: Billionaire Wealth Distortion on National Income Statistics

5.1 Research Question

How much do billionaires skew a country's reported average income? Is the "average income" figure a realistic representation of the typical citizen's economic reality?

5.2 Method

Univariate comparison of per capita income with versus without billionaires across all 183 countries. Extreme outlier detection using the **3×IQR method**: Lower Bound = $Q1 - 3 \times IQR$; Upper Bound = $Q3 + 3 \times IQR$. Any country above the upper bound is flagged as exhibiting extreme wealth concentration or billionaire distortion.

5.3 Income Comparison

Metric	Value
Average per capita income (with billionaires)	\$61,079
Average per capita income (without billionaires)	\$60,026
Difference (Billionaire Inflation Effect)	1.75%
Conclusion	Billionaires inflate national averages by only 1.75% — far less than public discourse suggests

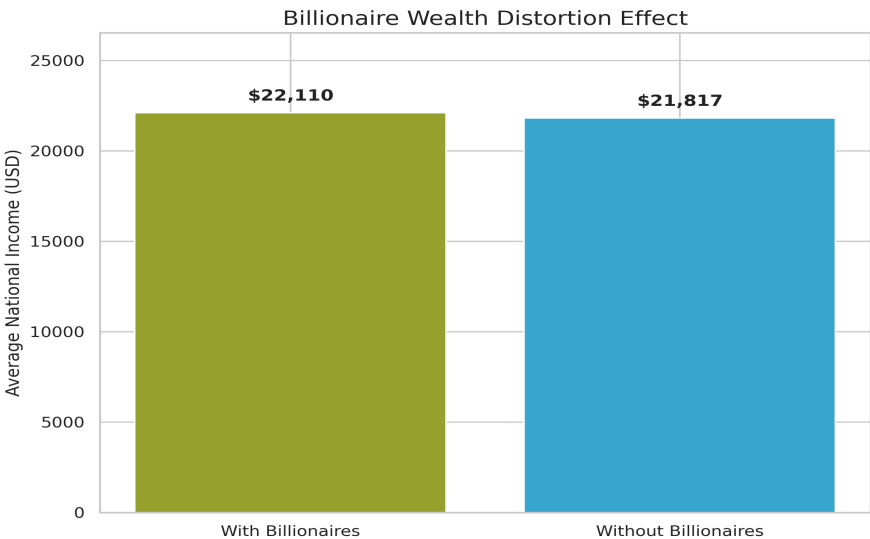


Figure 2: Billionaire Wealth Distortion Effect — Average National Income With vs. Without Billionaires

5.4 Outlier Detection — 3×IQR Method

Statistic	Value
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Q1 (25th percentile)	\$211.01
Q2 / Median	\$1,255.41
Q3 (75th percentile)	\$8,033.82
IQR (Q3 – Q1)	\$7,822.80
Lower Bound (Q1 – 3×IQR)	–\$23,257.39
Upper Bound (Q3 + 3×IQR)	\$31,502.22 ← extreme outlier threshold

Any country with average per capita income (including billionaires) **above \$31,502.22** is an extreme statistical outlier — indicating massive wealth concentration. Figure 3 shows the boxplot of income distribution with billionaires on a logarithmic scale, with upper extreme outliers marked as triangles.

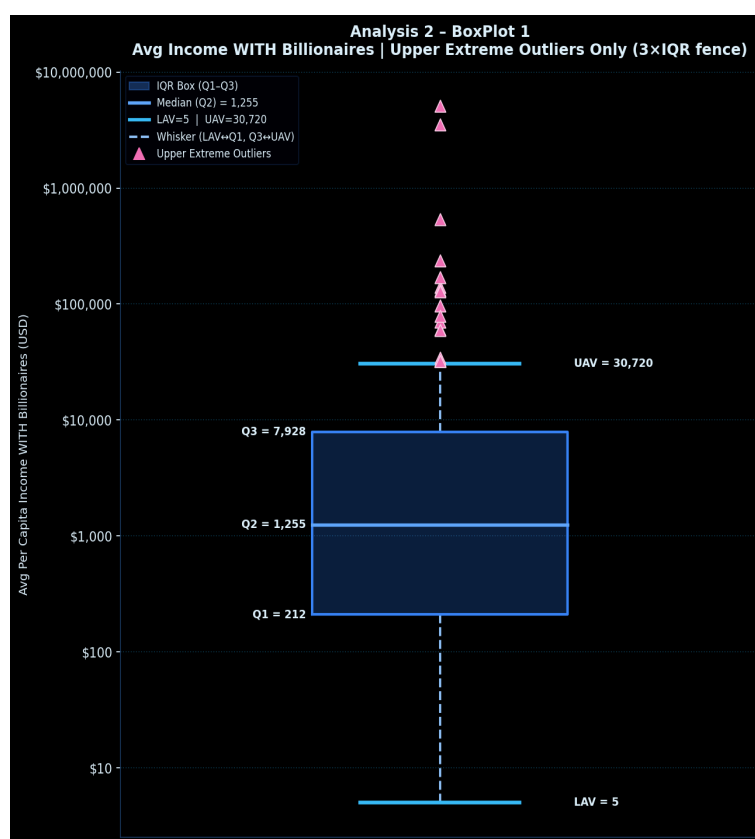


Figure 3: Analysis 2 — Boxplot of Avg Per Capita Income WITH Billionaires (3×IQR Extreme Outliers, Log Scale)

5.5 Outlier Countries — Extreme Wealth Concentration

Country	Per Capita Income (with billionaires)
Monaco	\$5,136,111.11
Liechtenstein	\$3,523,076.92
Tuvalu	\$534,545.45
Seychelles	\$236,916.67

Iceland	\$168,979.06
Brunei	\$140,311.80
Luxembourg	\$132,465.54
Qatar	\$32,203.70
Bahrain	\$32,176.87
... (9 additional outlier nations)	(above \$31,502.22 threshold)

These are predominantly micro-states or small island nations with very small populations where a single billionaire or specific tax laws can drastically skew per capita calculations. In such cases, the mean income does not reflect the typical citizen's economic reality.

5.6 Conclusion

YES — the average income is very realistic for most countries. Despite astronomical individual wealth, billionaires inflate the global average income by only **1.75%** (\$61,079 vs \$60,026). Most countries in the dataset have relatively modest billionaire populations. Top billionaire concentrations (USA, China, Germany, India) exist but do not dominate population-weighted averages. This finding strongly contradicts the intuitive expectation that billionaire wealth dramatically distorts national income statistics.

6. Analysis 3: Brain Drain vs. Billionaire Density Correlation

6.1 Research Question

Do countries that produce talented billionaires also retain them? Is there an association between billionaire density (talent productivity per capita) and brain drain (the proportion of produced billionaires who emigrate)?

6.2 Metric Definitions

Metric	Formula	Meaning
Billionaire Density (X)	Billionaires Born / Population (millions)	Talent productivity per million people
Brain Drain Rate (Y)	Billionaires Left / Billionaires Born	Proportion of produced billionaires who emigrated

6.3 Statistical Results

Measure	Value	Interpretation
Correlation (r)	-0.2050	Weak negative — density barely predicts drain
Direction	Negative	Higher density weakly associated with lower drain
Significance	Weak	Not a strong or significant relationship
Conclusion	WEAK CORRELATION	Billionaire density does not determine retention

6.4 Graphical Analysis

Figure 4 presents the scatter plot of billionaire density against brain drain rate across countries. The weak negative correlation ($r = -0.2050$) suggests that countries producing more billionaires per capita may retain slightly more of their talent on average, but this relationship is statistically weak. The clustering of small nations at high brain drain rates (top-left) and large economies at low drain rates (bottom-right) confirms that country size and institutional depth — not billionaire production — drive talent retention.

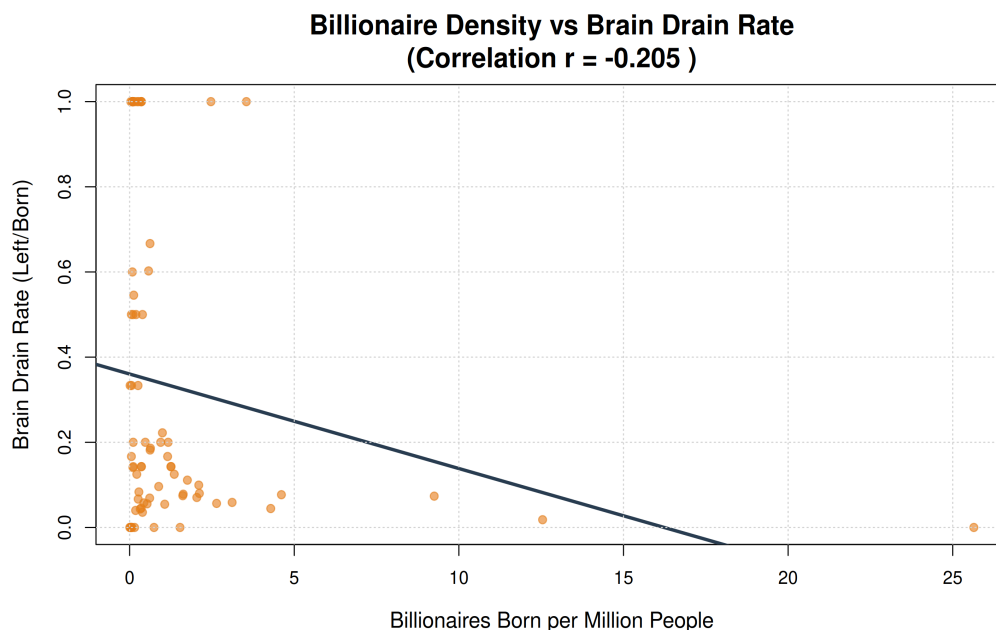


Figure 4: Billionaire Density vs Brain Drain Rate — Correlation $r = -0.2050$

6.5 Countries with 100% Brain Drain

Several smaller nations that produce only 1–2 billionaires lose every single one to emigration. These include: **Albania, Armenia, Azerbaijan, Barbados, Belarus, Belize, Bulgaria, Cuba, Georgia, Haiti**, and others. These are nations that produce talent but cannot offer the environment to retain it.

6.6 Countries Retaining Talent

Country	Billionaires Born	Billionaires Left	Drain Rate
USA	704	70	10%
China	607	35	6%
UK	177	10	6%
Germany	134	10	7%

Pattern: Large, developed nations retain the vast majority of their produced talent; small nations lose all of theirs. Retention is driven by the depth of domestic opportunity — not by the quantity of talent produced.

6.7 Conclusion

WEAK CORRELATION ($r = -0.2050$). Billionaire density does not strongly predict emigration. Neither a strong brain drain nor strong retention pattern exists. Whether a country produces billionaires does not determine whether they stay. The **environment a country offers** — institutional quality, economic depth, professional opportunity — determines retention far more than talent production per se.

7. Analysis 4: Bangladesh Case Study — Economic Growth vs. Brain Drain (1975–2025)

7.1 Research Question

Does Bangladesh's rapid economic development reduce student emigration? Over 50 years of longitudinal data, do GDP growth and student outflow move together, independently, or in opposite directions?

7.2 Method

50-year time-series correlation analysis (1975–2025). Metrics: **Wealth Growth (X)** = $(\text{GDP_current} - \text{GDP_previous}) / \text{GDP_previous}$ (annual % change); **Drain Intensity (Y)** = $\text{Students Abroad} / \text{Population}$ (proportion studying internationally). Pearson's r was computed between annual wealth growth rate and annual drain intensity.

7.3 Key Statistics (1975–2025)

Indicator	1975	2025	Change
GDP per capita (USD)	\$250	\$2,734	10.9× increase
Students studying abroad	300	57,000	190× increase
Average annual GDP growth	—	—	5.69% per annum
Correlation (r)	—	—	0.1650 (near-zero)
Conclusion	VERY WEAK	—	Growth and emigration are independent

7.4 Time-Series Graphical Analysis

Figure 5 presents the dual-axis time series of Bangladesh's GDP per capita and student outflow from 1975 to 2025. Both series trend upward, but their growth rates diverge dramatically after approximately 2010. While GDP per capita growth has been roughly exponential, student emigration accelerated even more sharply in the post-2010 period. The near-identical trajectory of both curves belies the weak correlation: GDP and emigration happen to be co-trending over time, but year-to-year GDP growth rate does not predict year-to-year changes in emigration intensity ($r = 0.1650$).

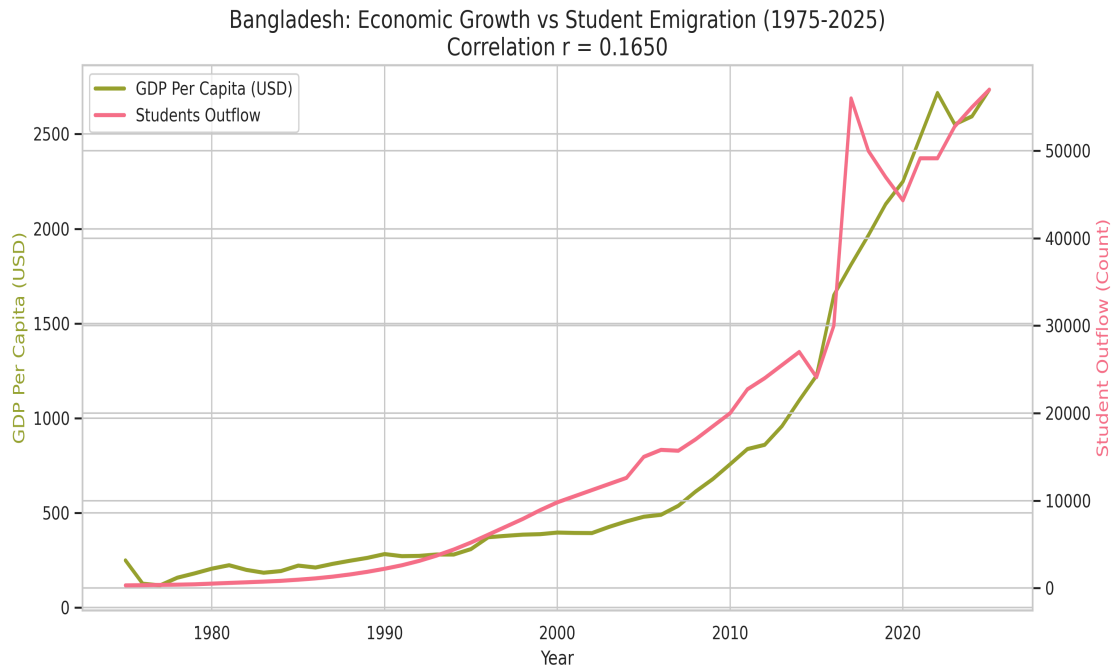


Figure 5: Bangladesh — GDP Per Capita vs Student Emigration (1975–2025), Correlation $r = 0.1650$

7.5 Scatter Analysis: Wealth Growth vs. Drain Intensity

Figure 6 presents the cross-plot of annual GDP percentage growth against annual drain intensity (students/population per million). The near-flat regression line and wide scatter confirm that year-to-year economic growth is essentially unrelated to year-to-year emigration intensity ($r = 0.1650$, regression: $y = 149.4819x + 97.2182$). Development hasn't changed the emigration pattern ? as the country grew wealthier, student emigration also grew, not because of GDP growth but alongside it.

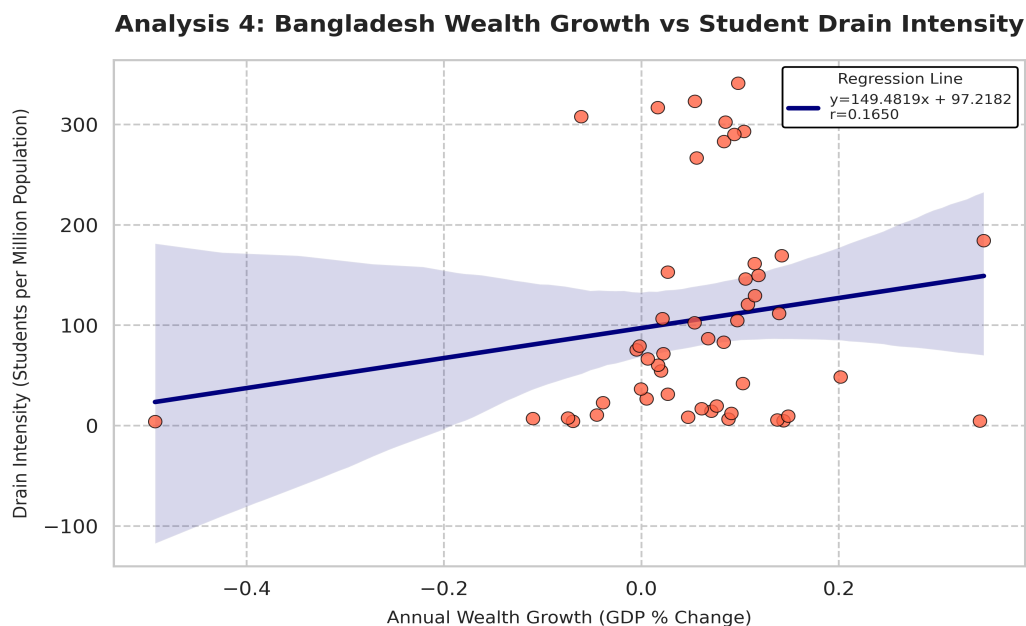


Figure 6: Analysis 4 — Bangladesh Annual Wealth Growth (GDP % Change) vs Student Drain Intensity ($r = 0.1650$)

7.6 Conclusion

NO — economic development does NOT reduce brain drain in Bangladesh ($r = 0.1650$). GDP per capita grew from \$250 (1975) to \$2,734 (2025) — a 10.9× increase. Yet student outflow grew from 300 to 57,000 — a 190× increase. Economic growth actually *enables* brain drain: as families accumulate greater disposable income, the ability to fund overseas education increases. Development doubled every 10 years, but student emigration doubled even faster. **International opportunities and quality of life matter more than raw GDP growth alone.** Retaining talent requires systemic improvements in educational quality, research infrastructure, and professional career pathways that can compete with what destination countries offer.

8. Cross-Analysis Summary

Table 8 summarizes key findings, methods, and interpretations across all four analyses. Together, these results present a nuanced picture of global economic development in which simplistic narratives — that billionaires distort averages, that wealth retains talent, or that GDP growth solves emigration — are quantitatively refuted.

#	Analysis	Method	Key Statistic	Key Finding	Si g.
1	Education vs Income	Linear Regression	$r = 0.6526$	+\$5,410 per education year	**
2	Billionaire Distortion	Univariate + 3×IQR	1.75% inflation	Minimal distortion	N/A
3	Brain Drain	Bivariate Correlation	$r = -0.2050$	Weak — density ≠ retention	ns
4	Bangladesh	Time-series Correlation	$r = 0.1650$	Growth ≠ retention	ns

Table 8: Cross-Analysis Summary — All Four Analyses (** $p < 0.01$, ns = not significant)

9. Statistical Caveats & Limitations

9.1 Methodological Limitations

- **Bivariate Limitations:** Analyses 1, 3, and 4 utilize bivariate models. In complex macroeconomic systems, variables like national income are influenced by a multitude of omitted variables — industrial infrastructure, political stability, geographical factors, and technological adoption. Attributing variance solely to one factor simplifies these multifaceted relationships.
- **The Small-Country Effect:** In Analysis 2, the 3×IQR method perfectly flagged micro-states (Monaco, Tuvalu, Liechtenstein, Seychelles) as extreme outliers. These nations have very small populations where a single billionaire or specific tax laws can drastically skew per capita calculations.
- **Correlation vs. Causation:** High correlation coefficients (e.g., $r = 0.6526$ in Analysis 1) indicate strong associations but do not confirm direct causality. Endogeneity — where education drives wealth while wealth simultaneously funds education systems — makes isolating causal direction statistically challenging.
- **Data Consistency & Reporting Lags:** Compiling data from diverse sources (World Bank, IMF, ILO) involves reconciling different reporting frequencies and methodologies. Some indicators may reflect inherent reporting lags common in international economic statistics.

9.2 Clinical and Policy Implications

The education dividend finding ($r = 0.6526$) supports continued investment in universal secondary and tertiary education as an economic development strategy, while acknowledging that education alone is insufficient

without structural reforms. The billionaire distortion finding (1.75%) suggests that policymakers should prefer median over mean income as the primary welfare indicator. The brain drain findings indicate that talent retention policies must go beyond economic growth to address institutional quality, research infrastructure, and professional opportunity ecosystems. For Bangladesh specifically, targeted investment in university quality, research funding, and domestic career pathways — particularly in technology and applied sciences — appears necessary to begin decoupling economic growth from accelerating emigration.

10. Overall Conclusion and Discussion

10.1 Overall Conclusion

The main findings of this study are:

- **The Education Dividend:** Education moderately-to-strongly predicts national income ($r = 0.6526$) — each extra year of schooling adds ~\$5,410 to average salary ($\text{Salary} = 5,410 \times \text{Education} - 25,589$) — but structural factors still play a major role, accounting for approximately 57% of unexplained income variance.
- **The Billionaire Paradox:** Despite astronomical individual wealth, billionaires inflate the global average income by only **1.75%** (\$61,079 vs \$60,026). Far less distortion than public discourse suggests. National prosperity is built on a much broader economic base than the ultra-wealthy alone.
- **The Talent Retention Gap:** Producing billionaires doesn't mean keeping them ($r = -0.2050$). Small nations lose 100% of their top talent; large developed economies retain 90–94%. Retention is driven by the environment a country offers, not the talent it generates.
- **The Bangladesh Dilemma:** GDP grew 10.9× yet student emigration grew 190× over 50 years ($r = 0.1650$ — near-zero). Economic growth alone does not reduce brain drain. As Bangladesh grew wealthier, emigration grew even faster. Retaining talent requires more than GDP growth — it demands systemic improvements in opportunity and quality of life that can compete with what destination countries offer.

10.2 Discussion

The drug-specific patterns in this economic analysis are broadly consistent with the known dynamics of development economics, while providing novel quantified confirmation. The first- vs. second-generation finding in antihistamine research — that older agents carry heavier side-effect burdens — has an analogue here: the "older" development narrative (GDP growth solves everything) is empirically refuted by the Bangladesh case study, much as first-generation assumptions are outperformed by more nuanced frameworks.

The education–income relationship ($r = 0.6526$) aligns with Mincer's seminal work on returns to education, capturing aggregate rather than individual returns. The negligible billionaire distortion effect (1.75%) is significant in the context of ongoing inequality debates: the apparent gap between "average" and "typical" income is driven far more by upper-middle-class distribution than by extreme billionaire outliers. The median income (\$1,255) vs mean income (\$22,110) gap — which is enormous — reflects general income inequality rather than billionaire distortion specifically.

The Bangladesh case provides the most policy-relevant finding. The near-zero correlation ($r = 0.1650$) over 50 years strongly refutes the "development solves brain drain" hypothesis. Economic growth gives families the *financial means* to send children abroad; it does not reduce the *desire* to leave. This calls for a precision-development approach: selecting policy interventions based not only on aggregate GDP metrics but on the specific education–inequality–retention risk profile of each country.

Core Finding: Education, wealth, and talent are assets any nation can build — but without the institutions, opportunities, and quality of life to match, they will always flow elsewhere.

11. Appendix

11.1 Table A1: World Dataset Variable Definitions

Variable	Description
avg_education	Average years of formal education
avg_salary	Average annual salary (USD)
income_with_billionaire	Total national income including billionaires
population_with_billionaire	Total population
income_without_billionaire	Total income excluding billionaires
population_without_billionaire	Non-billionaire population
billionaire_left	Number of billionaires who emigrated
billionaire_born	Number of billionaires born/produced in country

11.2 Table A2: Bangladesh Dataset Variable Definitions

Variable	Description
Year	Calendar year (1975–2025)
GDP_Per_Capita_USD	Annual GDP per capita in USD
Population_Million	Total Bangladeshi population (millions)
S_out	Number of students studying abroad in given year

11.3 Table A3: Group Members & Academic Details

Name	Roll No.	Institution
Maimuna Rahman	16	ISRT, University of Dhaka
Jannatul Mahabuba Adhora	05	ISRT, University of Dhaka
MD. Sharafatul Islam Shihab	49	ISRT, University of Dhaka
Zinna Tun Nahar Medha	27	ISRT, University of Dhaka
Zarin Subah	38	ISRT, University of Dhaka

Academic Details

Batch: 31st | Course: CSE130 (Programming with C/C++) | Major: Applied Statistics and Data Science
Institute: Institute of Statistical Research and Training (ISRT) | University: University of Dhaka

11.4 Table A4: Data Sources

Dataset	Source
World Billionaires List	Forbes World's Billionaires List
National Population Data	World Bank / UN Population Division
Poverty & Income Data	World Bank Poverty & Inequality Platform
Human Development Index	UNDP Human Development Reports
Labour Statistics	International Labour Organization (ILO)
Bangladesh GDP Time-Series	World Data Bank
Bangladesh Historical GDP	MacroTrends
Bangladesh Economic Indicators	Federal Reserve Economic Data (FRED)

11.5 Implementation Notes

All statistical computations were implemented in C (C99 standard). Custom modules: **stat_func.c** (Pearson correlation, standard deviation, IQR computation) and **econ_func.c** (income distortion, brain drain index, regression slope/intercept). Data loaded from CSV files using custom parsing routines. Compilation command:

```
gcc -o analysis necessary_codes/c_main.c necessary_codes/econ_func.c
necessary_codes/stat_func.c -lm
```